AMCAT Guide

The Aspiring Minds Computer Adaptive Test (AMCAT) is an adaptive computer test and measures the employability skills of a job candidate. Over 700 companies have partnered with AMCAT to use the scores to evaluate the employability of job candidates. It helps recruiters assess candidates on critical skills and domain knowledge. The AMCAT test is taken by students from all streams including Engineering, Management, Arts, and Commerce.

Below are the details of modules offered by AMCAT to a particular student and its duration:

Modules offered in AMCAT_BE, M.E., M.Sc. Data Science & MCA		
Stream	**Modules stream-wise**	**Duration**
CSE/IT/MCA	English; Logical; Quant; Personality; Computer Programming; Automata; Automata Fix; Computer Science, WriteX	193 minutes
ECE	English; Logical; Quant; Personality; Computer Programming; Automata; Automata Fix; Electronics and Semi-Conductors, WriteX	188 minutes
Mechanical/Automobile	English; Logical; Quant; Personality; Computer Programming; Mechanical Engineering; Automobile Engineering, WriteX	138 minutes
Petroleum	English; Logical; Quant; Personality; Computer Programming; Petroleum Engineering, WriteX	123 minutes
Chemical	English; Logical; Quant; Personality; Computer Programming; Chemical Engineering, WriteX	123 minutes
Civil	English; Logical; Quant; Personality; Computer Programming; Civil Engineering, WriteX	128 minutes
Mechatronics	English; Logical; Quant; Personality; Computer Programming: Mechanical Engineering; Electronics & Semi-Conductors, WriteX	138 minutes
Aerospace	English; Logical; Quant; Personality; Computer Programming; Aerospace, WriteX	128 minutes
Electrical	English; Logical; Quant; Personality; Computer Programming; Electrical Engineering, WriteX	123 minutes
Biotech	English; Logical; Quant; Personality; Computer Programming ; Molecular Biology; Biochemistry, WriteX	133 minutes
Food Technology	English; Logical; Quant; Personality; Computer Programming ; Food Science, WriteX	123 minutes
M.Sc. Data Science	English; Logical; Quant; Personality; Data Science, Computer Programming; Automata, Automata Fix, Computer Science, WriteX	218 minutes

Time bifurcation of each Module and No. of Questions			
Module	**Total Questions**	**Min count of questions to Attempt**	**Duration of each section**
English	18	14	18 Min
Logical	12	8	15 Min
Quant	16	14	20 Min
Personality	72	No Minimum Question Criteria	15 Min
Computer Programming	12	5	15 Min
Automata	2	No Minimum Question Criteria	45 Min
Automata Fix	7	No Minimum Question Criteria	20 Min
Computer Science	20	8	20 Min
Electronics & Semi-Conductors	12	6	15 Min
Mechanical Engineering	12	6	15 Min
Petroleum Engineering	20	No Minimum Question Criteria	15 Min
Chemical Engineering	12	6	15 Min
Civil Engineering	20	No Minimum Question Criteria	20 Min
Electrical Engineering	12	5	15 Min
Automotive Engineering	15	6	15 Min
Aerospace Engineering	18	No Minimum Question Criteria	20 Min
Molecular Biology	20	8	15 Min
Biochemistry	12	5	10 Min
Food Science	18	14	15 Min
Data Science	12	No Minimum Question Criteria	25 Min
Write X	1	No Minimum Question Criteria	25 Min

AMCAT will be conducted with mandatory live video stream option under the supervision of vigilant invigilators. It is imperative to have good internet connectivity with minimum 2MBPS bandwidth and a laptop /PC with working mic and webcam so that video streaming is smooth.

Please ensure the compliance to the following to appear for the test:

A. **System Requirements**:

- A personal computer / Laptop (The test will not work on any mobile phones)
- A webcam connected with the computer/In-built webcam in the laptop
- **Smooth and uninterrupted internet connectivity (throughout the testing session) with minimum 2 MBPS bandwidth**
- **Working microphone of the personal computer/ laptop.**

Note:-Simple earphones may be connected in case the microphone is not working or not available but earphones **should NOT be plugged into the ears** otherwise Live Proctoring may disqualify you).

B. **Browser Requirements**:

- Google Chrome (version 40 and above)
- Mozilla Firefox (version 40 and above)
  Note- In case the test is taken on Windows 8, the latest version should be used and all the popups/ notifications should be disabled.

C. Important Guidelines:

- Please note down the AMCAT ID in your records for future reference.
- Do not copy-paste the User Name and Password and only **type** the same.
- Complete the test in one go as no breaks are allowed in-between.
- Keep a track of time as the test is timed and will be auto-submitted after the completion of the time.
- Mandatorily answer all the questions while completing all the modules.
- **You can't skip any question or not go back to the previous question.**
- **There is no negative marking.**
- If somehow the test is stopped in between, (due to power failure or internet disconnection) wait for five minutes and re-login using the same details. After entering the username and password, click on 'Resume Incomplete Test', select your AMCAT Id and then re-enter Username and Password.
- **Ensure that you resume Test within 2 hours of your test being stopped.**
- Ensure that you are not wearing any headphones, earphones, etc.
- Do not cover your face during the test.
- Ensure that no one else is sitting with you during the entire duration of the test.

D. Procedure:

- Type the URL (**https://amcatcampus.aspiringminds.com**)in the address bar of the Web Browser.
- Enter the username and password shared via auto-generated email sent by Aspiring Minds.
- Next, click on 'Start New Test' and note your Test Id as mentioned on the top right corner of the screen.
- You will be automatically prompted to 'Allow' your Camera to be enabled, please 'ACCEPT' the request. This will prompt a message 'You Are Under Surveillance'. Your camera is now enabled.
- Give permission to access your camera through the automatically generated pop-up.
- Once you ACCEPT the request, there will be a pop-up message 'You are Under Surveillance' which confirms that camera is now enabled.
- To answer a question, click on Confirm and then Next to submit your answer.
- Do not click on Module Exit or Test Exit.
- If your test gets stuck at any point, there is a 'Resume Test' option that will enable you to restart the test at the same place where you got stuck. Please ensure that you Resume the Test within 2 hours of your test being stopped, otherwise your login will expire, and you won't be able to complete the test.

E. Do's & Don't- Instructions:

- Complete the test in one go, no breaks allowed in-between.
- This is a timed test; it is essential to keep track of time while answering the questions.
- Answer all the questions and complete all the modules.
- There is no negative marking.
- You cannot change your answer to any previous question.
- Do not click on Module Exit or Test Exit.
- Ensure that while taking the test, no other tab/window/link is opened simultaneously on your system else your scores will not be processed.

- Do not try to take screenshots of the test, else your test will be locked & you won't be able to complete the test.
- Please disable all POP-UPS/ Notifications before starting the test.
- Ensure your face is clearly visible on the webcam. Do not wear a cap, scarf, goggles/ sunglasses, headphones, earphones, etc., or hide your face during the test.
- Please ensure that no one else is sitting with you during the entire duration of the test.

Note- No candidate will be allowed to login to the test after 20 minutes after the start of the scheduled time of the test. In such a case, the candidate will be marked absent.

There may be instances when the test is auto-submitted or gets locked. The reasons for the same are outlined below:

- Switching through tabs which leads to locking of the test and test cannot be resumed again.
- Taking screenshots of the test.
- Pop-ups during the test.
- Not sitting in front of your computer during **entire duration** of the test.

Note- It is mandatory for a student to be in front of the camera for the entire duration of the test, else the test will be ENDED by CU invigilators IMMEDIATELY.

Other reasons which will attract <u>IMMEDIATE END of the test by CU invigilators</u> are-

- **If a student tries to cover his/her camera or a disturbance is witnessed in the camera during the test.**
- **If a student is involved in any unethical activity related to the test or is found cheating by any means.**
- **In case of cheating or absenteeism, student will be debarred from TPP and will not be allowed to sit for ANY placement drives.**

Some of the actions will can account to cheating are listed below-

- Using mobile phones during the test.
- Moving away from the camera
- Not sitting in the center of the screen
- Taking screenshots of the test via any means such as screen mirroring, through cell phones, etc.
- Talking with someone during the test
- Wearing earphones
- Some other person walking behind the student
- Some other person looking at the screen
- Use of calculators
- Wearing masks or covering face with any piece of cloth
- Looking here and there during the test
- Presence of people other than the test taker in the test area

<u>Importance of Modules offered by AMCAT:</u>

The AMCAT Exam tests the student's knowledge in compulsory modules like– English, Quantitative Ability, Logical Ability and Personality Test.

- **<u>English</u>** - Familiarity with the English Language in its various nuances is an essential skill, especially in the current climate of global networking. Ideally, any recruitment should involve a test of skills in handling the language in ways

that promote the objectives of the company. Needless to state, an appropriate test is necessary. English test uses a variety of internationally standardized resources for framing questions aimed at determining the candidate's ability to

(a) Understand the written text

(b) Comprehend the spoken word and

(c) Communicate effectively through written documents.

- **Quantitative Aptitude-** The Quantitative Ability assesses the ability of the candidate in following two aspects:

  (a) Basic understanding of numbers and applications This section tests whether the candidate has understanding of basic number system, i.e., fractions, decimals, negative, positive, odd, even numbers, rational numbers, etc. The candidate should know how to do basic operations on these numbers, understand concepts of factors/divisibility and have good practice of algebra. Apart from operations on numbers, the candidate should know how to converta real-world problem into equations, which is to be solved to find an unknown quantity. The candidate is tested on Word Problems representing various scenarios to assess the same.

  b. Analytical Maths - These are aspects of mathematics needed for Engineering disciplines and data analysis. This includes permutation-combination, probability and understanding of logarithms.

- **Logical Ability -** The Logical Ability section assesses the capacity of an individual to interpret things objectively, to be able to perceive and interpret trends to make generalizations and be able to analyze assumptions behind an argument/statement. These abilities are primary for success of a candidate in the industry. Specifically, these are divided into following sections:

  a. Deductive Reasoning: Assesses the ability to synthesize information and derive conclusions.

  b. Inductive Reasoning: Assesses the ability to learn by example, imitation or hit-andtrial. This alsoprovides an indication of how creative the individual is.

  c. Subjective Reasoning: Assesses the critical thinking ability of an individual to see through loopholes in an argument or group of statements. All these abilities are tested both using numerical and verbal stimuli. Coachable questions have been identified and removed.

- **Computer Programming** - The Computer Programming Principles module evaluates the suitability of the candidate for the software industry. It not only tests the knowledge and application of basic constructs of programming, but also concepts of data structures, algorithm analysis and object-oriented-programming. The test is language-independent and all programming questions use a pseudo-code. Significant effort has been made to exclude memory-based and rote-learning questions. The test contains questions on debugging programs, finding the output of programs, completing incomplete programs, finding complexity of algorithms, questions on implementation and operations on different data structures, etc.

- **Electronics and Semiconductor Engineering** - The Electronics and Semiconductor test assesses the suitability of the candidate for the SOC, Embedded Systems, VLSI design, etc. companies. This test together with that of Computer Programming assesses the suitability of candidates for EDA companies. The test has

the following sections:

a. Analog Electronics

1. Basic Components, their operations and Circuit Analysis
2. Active Components, Large, Small Signal and Circuit Analysis
3. Frequency domain and time domain analysis of systems, Feedback and Stability
4. Opamp based circuits and analysis

b. Digital Electronics

1. Boolean Algebra, Minimization of Boolean Functions
2. Implementation and Analysis of logic gates
3. Sequential blocks - flip-flops and latches
4. Digital Circuits and Blocks 5. State Machines and design of Complex sequential circuits

- **Mechanical Engineering** - In this module, a student is tested for his understanding of mechanical engineering - theoretical and practical knowledge. Questions from different areas in this subject are asked so as to assess a student on his complete knowledge of the subject. The test has the following sections:

  a. Manufacturing Science

  b. Thermodynamics & IC Engines

  c. Fluid and Machine Mechanics

- **Electrical Engineering** - The Electrical Engineering module has been designed to assess a candidate's knowledge working in power sector. The module is meant for B Tech. students who may be freshers or the students who may be exposed to industry for one to two years. The module checks for the concepts which would be used by the engineers in everyday working. The module consists of both conceptual and practical aspects of the subject.

- **Civil Engineering** - Civil Engineering module assesses a student's skills, knowledge and understanding of the core ideas involved in the branch of civil engineering. The module focuses on testing a student on theoretical knowledge and practical concepts which will help him perform a good job as an engineer in the industry.

- **Automata** - Automata is a path-breaking tool to evaluate programming skills in a simulated environment. With a capability to evaluate programming code beyond correctness and ability to rate code quality, efficiency of execution and to provide objective norm based scores, Automata is a leap ahead of commonly available tools in the genre.

The assessment allows evaluation of actual programming skills of a candidate, giving the candidate an opportunity to write the program in an editor, compile and run test cases, all in the assessment environment itself.

Automata come with a scientifically designed question bank consisting of a wide range of easy-to-hard programming problems with smartly designed corner and generic tests cases. It evaluates a candidate's knowledge of algorithms, data structures, concepts like recursion, dynamic memory allocation and modular programming. The compiler supports multiple programming languages, such as C, C++, Java, etc. Some of the unique features of Automata include:

- With a variety of basic, advanced and edge test cases designed for a problem, the code is checked thoroughly for correctness and completeness
- It is capable of evaluating the time complexity or simply the efficiency of the code
- The code is scored on quality of the code based on industry defined best practice
- A detailed report with objective and subjective evaluation of the code

Automata can be taken alongside AMCAT, Aspiring Minds' standardized online, adaptive assessment test. It can be taken at a campus event or at our test centers across the country. Detailed automated reporting on real time and features like interviewer feedback make it easy to use in a mass scale. Automata Pro can effectively help in entry-level jobs as well as experienced programming jobs.

- **Data Science** - The Data Science module will assess your knowledge of Probability and Statistics, ML Algorithms, and Implementation and setting up of ML Experiments.

- **AMPI: Aspiring Minds Personality Inventory** - It is the first personality inventory designed for personality analysis of Indian college graduates for the purpose of inputs to corporate personnel selection. AMPI is based on the five factor model, which is by far the only scientifically validated and reliable personality model. Several scientific studies across the world have shown that different combinations of the five factor personality traits strongly correlate to different job profiles and predict long term job performance reliably. AMPI analysis will be a worthwhile objective input to the corporate selection process and help find better matches to job profiles. The AMPI questionnaire asks for candidate's reaction under various scenarios, his/her beliefs, likes dislikes to ascertain his/her personality factors. Factors map to traits such as candidate motivation, self-discipline, sociability, persistence, confidence, emotional stability, etc. which both intuitively and scientifically map to job requirements. AMPI builds in a strong proprietary methodology to control distortions due to social desirability and answer faking. AMPI has been designed specifically keeping the fresh Indian graduates in mind. Context is very important in design of items. AMPI items take into consideration the cultural sensibilities of Indians, the scenarios students face at college/home, also depending on the socio-economic status of the target population. This brings AMPI into a unique position as compared to generic/Western inventories, which do not suit our target population and fail miserably. AMPI's scoring is based on statistical techniques of factor analysis, polytomous item analysis and structural modeling. Norms have been set on large candidate assessment done on final year graduates. Test forms are auto-generated such that each factor can be reliably predicted in feasible amount of time. Test-retest reliability and test validity are statistically guaranteed.

AMPI traits are:

a. Extraversion b. Conscientiousness c. Emotional Stability

d. Openness to Experience e. Agreeableness